Tutorial 3: First-Order ODEs (Part III)

[Attempt all questions before tutorial session]

A) Identify the type of a first-order ODE

Determine all values of the constants m and n, if there are any, for which the DE

$$\left(x^5 + y^m\right)dx - x^n y^3 dy = 0 \tag{1}$$

is of each of the following.

(a) Exact, (b) Separable, (c) Homogenous, (d) Linear, (e) Bernoulli

Key:

$$\text{The equation (1) is} \begin{cases} \text{exact,} & \text{if } \; m = n = 0, \\ \text{separable,} & \text{if } \; m = 0, \; n \in \mathbb{R}, \\ \text{homogenous,} & \text{if } \; m = 5, \; n = 2, \\ \text{nonlinear,} & \text{for any } m, n, \\ \text{Bernoulli,} & \text{if } \; m = 4, \; n \in \mathbb{R}. \end{cases}$$

Solution: The given DE

$$\left(x^5 + y^m\right)dx - x^n y^3 dy = 0 \quad \Longleftrightarrow \quad \frac{dy}{dx} = \frac{x^5 + y^m}{x^n y^3}. \tag{2}$$

(a) To be exact, we requite that

$$P_y = my^{m-1} = Q_x = -nx^{n-1}y^3 \tag{3}$$

Comparing the powers of x, y, we have to take $n = 1$ and $m = 4$, but with these choices, the equation (3) does not hold. Therefore, the given DE (1) can not be an exact DE for any m, n, except for $m = n = 0$.

(b) We now check for what values of m, n, the equation (1) can be separable. We rewrite the equation as

$$y^3 dy = \frac{x^5 + y^m}{x^n} dx.$$

We see that for $m = 0$ and any real number n, the equation is separable.

(c) To examine whether the equation could be homogenous, we denote

$$\frac{dy}{dx} = \frac{x^5 + y^m}{x^n y^3} = f(x, y),$$

and find that

$$f(tx, ty) = \frac{(tx)^5 + (ty)^m}{(tx)^n (ty)^3} = \frac{t^5 x^5 + t^m y^m}{t^{n+3} x^n y^3}$$

which equals to $f(x, y)$, i.e., independent of t, when

$$m = 5, \quad n + 3 = 5 \implies n = 2, \quad m = 5.$$

Accordingly, if $n = 2$ and $m = 5$, the given DE is homogenous.

(d) To see when the equation is a linear DE, we rewrite (2) into a standard form

$$\frac{dy}{dx} - \frac{1}{x^n} y^{m-3} = -x^{5-n} y^{-3}. \tag{4}$$

This equation can not be linear due to the term y^{-3}.

(e) We find from (4) that for $m - 3 = 1$ and any real number n, the equation is a Bernoulli's DE of order -3.

B) **Riccati equation**

The Riccati equation takes the form

$$y' = p(x)y^2 + q(x)y + r(x), \tag{5}$$

where p, q and r are given continuous functions. Notice that when $r(x) \equiv 0$, this equation is reduced to the Bernoulli's equation of order $n = 2$. However, if $r(x) \neq 0$, this DE is difficult to solve. French Mathematician Liouville pointed out in 1841 that if we know one particular solution $\tilde{y}$ of (5), then through a change of variable $y = z + \tilde{y}$, we are able to convert the equation to the Bernoulli's equation.

(i) Prove this Liouville's statement.

(ii) Find the general solution of the Riccati equation:

(a) $\quad y' e^{-x} + y^2 - 2ye^x = 1 - e^{2x} \quad$ with $\quad \tilde{y}(x) = e^x$.

(b) $\quad x^2(y' + y^2) = 2 \quad$ with $\quad \tilde{y}(x) = -\dfrac{1}{x}$.

Solution: (i) Plugging $y = z + \tilde{y}$ into (5) leads to

$$\begin{aligned} & z' + \tilde{y}' = p(z+\tilde{y})^2 + q(z+\tilde{y}) + r \\ \Rightarrow & z' + \tilde{y}' = pz^2 + 2pz\tilde{y} + qz + \left(p\tilde{y}^2 + q\tilde{y} + r\right). \end{aligned}$$

Since $\tilde{y}$ is a solution to (5), i,e, $\tilde{y}' = p\tilde{y}^2 + q\tilde{y} + r$, we have

$$z' = pz^2 + 2pz\tilde{y} + qz.$$

Rewrite it as

$$z' - (2p\tilde{y} + q)z = pz^2$$

which is a Bernoulli equation of order $n = 2$, with

$$P(x) = -(2p\widetilde{y} + q), \quad Q(x) = p(x).$$

Using the solution formula for Bernoulli's equation, we find

$$z^{1-n} = e^{-\int (1-n)P(x)dx}\left\{\int (1-n)Q(x)e^{\int (1-n)P(x)dx} + C\right\},$$

that is,

$$z^{-1} = e^{-\int (2p\tilde{y}+q)dx}\left\{-\int pe^{\int (2p\tilde{y}+q)dx} + C\right\} \tag{6}$$

Hence, the general solution to the Ricatti equation $y = z + \tilde{y}$ with z given by (6).

Solution: (ii) (a) $y = e^x + \dfrac{1}{c+e^x}$, (b): $xy = \dfrac{2x^3 - c}{3x^3 + c}$.

(a) We write the equation as the standard form

$$y' = -e^x y^2 + 2ye^{2x} + e^x\left(1 - e^{2x}\right).$$

Notice that

$$p(x) = -e^x, \quad q(x) = 2e^{2x}, \quad r(x) = e^x\left(1 - e^{2x}\right), \quad \tilde{y} = e^x.$$

Now, we compute

$$\int (2p\tilde{y} + q)dx = \int 0dx = 0.$$

Therefore

$$z^{-1} = c + \int e^x dx = c + e^x.$$

$$\Rightarrow z = \frac{1}{c + e^x}.$$

Therefore, the general solution is

$$y = e^x + \frac{1}{c + e^x}.$$

b) The standard form of the Ricatti equation is $y' = -y^2 + \frac{2}{x^2}$ with a particular solution $\tilde{y} = -\frac{1}{x}$. Assume that the general solution is $y = z - \frac{1}{x}$. Then z satisfies

$$z' - \frac{2}{x}z = -z^2.$$

which is a Bernoulli's equation. We know that z can be solved by

$$\begin{aligned} z^{-1} &= e^{-\int \frac{2}{x}dx}\left\{\int 1 \cdot e^{\int \frac{2}{x}dx}dx + c\right\} \\ &= \frac{1}{x^2}\left\{\int x^2 dx + c\right\} = \frac{1}{x^2}\left\{\frac{x^3}{3} + c\right\}. \end{aligned}$$

Hence, we have

$$z = \frac{3x^2}{x^3 + 3c}.$$

The general solution is

$$y = z - \frac{1}{x} = \frac{3x^2}{x^3 + 3c} - \frac{1}{x} = \frac{2x^3 - 3c}{x\left(x^3 + 3c\right)}.$$

We denote $3c$ by c_1, and rewrite the solution as

$$xy = \frac{2x^3 - c}{x^3 + c}.$$

C) **Solving First-order ODE by Substitution**

Solve the following DE by using suitable substitutions:

$$\begin{aligned} &(a) \quad y' + e^{y'} - x = 0. \quad \text{(Hint: use } u = y'\text{)} \\ &(b) \quad xy' - y = 2x^2y(y^2 - x^2). \quad \text{(Hint: use } u = x^2y\text{).} \\ &(c) \quad y' + \frac{y}{x} = e^{xy}. \quad \text{(Hint: use } u = e^{xy}\text{)} \end{aligned}$$

Solution: (a) I removed this question, due to its difficulty, while this type of question may be accessed during the CAs or finals.

$u = y' \Rightarrow u + e^u = x$. Now, by the Chain Rule,

$$u = \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} \Rightarrow \frac{dy}{du} = u\frac{dx}{du}$$

Since by $x = u + e^x$, we have $\frac{dx}{du} = 1 + e^u$, so

$$\begin{aligned}\Rightarrow \frac{dy}{du} =& u\left(1+e^u\right) \Rightarrow y = \int u\left(1+e^u\right)du\\ =& \frac{u^2}{2} + (u-1)e^u + c.\end{aligned}$$

The general solution is

$$\begin{cases} x = u + e^u \\ y = \dfrac{u^2}{2} + (u-1)e^u + C \end{cases}$$

(b) Let $u = x^2 y \Leftrightarrow y = \frac{u}{x^2}$. We now eliminate y :

$$\frac{du}{dx} = 2xy + x^2\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{x^2}\frac{du}{dx} - \frac{2y}{x} = \frac{1}{x^2}\frac{du}{dx} - \frac{2u}{x^3}$$

The equation becomes

$$\frac{1}{x}\frac{du}{dx} - \frac{3u}{x^2} = 2u\left(\frac{u^2}{x^4} - x^2\right) \Rightarrow \frac{du}{dx} + \left(2x^3 - \frac{3}{x}\right)u = \frac{2}{x^3}u^3$$

It is the Bernuoulli's DE of order $n = 3$. Solving the Bernoulli equation leads to

$$u^{-2} = \frac{1}{x^6}\left(1 + ce^{x^4}\right)$$

We transform the variable: $u = x^2 y$ back. The general solution is

$$x^2 - y^2 = cy^2 e^{x^4}.$$

(c) We make a change of variable

$$u = e^{xy} \Rightarrow \frac{du}{dx} = e^{xy}\left(y + x\frac{dy}{dx}\right) \Rightarrow y' + \frac{y}{x} = \frac{1}{xu}\frac{du}{dx}$$

Hence the equation becomes

$$\frac{1}{xu}\frac{du}{dx} = u \Rightarrow \frac{1}{u^2}\frac{du}{dx} = x$$

The solution

$$-\frac{1}{u} = \frac{x^2}{2} + C \Rightarrow -e^{-xy} = \frac{x^2}{2} + C$$

so the general solution is

$$\frac{x^2}{2} + e^{-xy} = C.$$

D) **Reducible Second-order ODEs**

Reduce the following second-order ODE to first-order ODE and the solve them:

$$(a) \quad xy'' + y' = 8x, \quad x > 0, \qquad (b) \quad yy'' + (y')^2 = yy'$$

Solution: (a): $y = 2x^2 + c_1 \ln x + c_2$ (b): $y^2 = C_1 e^x + C_2$

(a): As y is missing, we introduce $v = y'$ so $v' = y''$ and we convert the equation into

$$xv' + v = 8x \quad \Rightarrow \quad v' + \frac{1}{x}v = 8.$$

We solve this first-order linear equation with IF:

$$I(x) = e^{\int 1xdx} = x,$$

so we have

$$v(x) = \frac{1}{x}\Big\{\int 8xdx + c_1\Big\} = \frac{1}{x}\{4x^2 + c_1\}.$$

Then

$$y = \int v(x)dx + c_2 = \int \frac{1}{x}\{4x^2 + c_1\}dx + c_2 = y = 2x^2 + c_1 \ln x + c_2.$$

(b) As y is missing, we introduce $v = y'$ so

$$y'' = \frac{v}{dx} = \frac{dv}{dy}\frac{dy}{dx} = v\frac{dv}{dy}.$$

We convert the equation into

$$yv\frac{dv}{dy} + v^2 = yv.$$

If $v \neq 0$, we can simplify it into

$$y\frac{dv}{dy} + v = y \quad \Rightarrow \quad \frac{dv}{dy} + \frac{1}{y}v = 1.$$

We solve this first-order linear DE:

$$v(y) = \frac{y}{2} + \frac{c_1}{y},$$

so we have the separable DE

$$\begin{aligned}
y' = v = \frac{y}{2} + \frac{c_1}{y} = \frac{y^2 + 2c_1}{2y} \quad &\Rightarrow \quad \frac{2ydy}{y^2 + 2c_1} = dx \\
\Rightarrow \quad &\int \frac{2ydy}{y^2 + 2c_1} = \int 1dx + c_2 \\
\Rightarrow \quad &\int \frac{d(y^2 + 2c_1)}{y^2 + 2c_1} = \int 1dx + c_2 \\
\Rightarrow \quad &\ln |y^2 + 2c_1| = x + c_2 \quad \Rightarrow \quad y^2 + 2c_1 = \pm e^{c_2}e^x \\
\Rightarrow \quad &y^2 = C_1 e^x + C_2.
\end{aligned}$$

1 Summary of Techniques for first-order ODEs

The techniques that we have derived for solving first-order DE will always work, provided they are applied to the correct type of equation. The difficulty that is often encountered at this stage is not in actually applying the techniques, but rather in determining which type the equation falls into. We summarize below the six basic types of DEs and their solution techniques.

Table 1: `Basic solution techniques for` $y' = f(x, y)$.

Type	**Standard Form**	**Technique**
Separable	$p(y)dy = q(x)dx$	*Separate the variables and integrate directly*
Linear	$y' + p(x)y = q(x)$	*GS:* $y = e^{-\int p(x)dx}\left[\int q(x)e^{\int p(x)dx}dx + C\right]$
Bernoulli	$y' + p(x)y = q(x)y^n (n \neq 1)$	*GS:* $y^{1-n} = e^{-(1-n)\int p(x)dx}\left[(1-n)\int q(x)e^{(1-n)\int p(x)dx}dx + C\right]$
Exact	$M(x, y)dx + N(x, y)dy = 0$, *with* $M_y = N_x$	*GS:* $u(x, y) = C$, *where the potential function* $u(x, y)$ *is solved from the system:* $u_x = M$, $u_y = N$
Non-Exact *with IF*	$P(x, y)dx + Q(x, y)dy = 0$, *with* $P_y \neq Q_x$	*Case I: If* $\dfrac{P_y - Q_x}{Q} = A(x)$, *then IF:* $I(x) = e^{\int A(x)dx}$ *Case II: If* $-\dfrac{P_y - Q_x}{P} = B(y)$, *then IF:* $I(y) = e^{\int B(y)dy}$
Homogenous	$y' = f(x, y)$ *where* $f(tx, ty) = f(x, y)$, $\forall t > 0$	*Change variables:* $y = xV$ *and* *covert it into a separable equation*